

A CERTIFIED UNCONDITIONAL UPPER BOUND $\Lambda \leq 0.1875$ FOR THE DE BRUIJN–NEWMAN CONSTANT

MOSAIC INTELLIGENCE

ABSTRACT. Let Λ denote the de Bruijn–Newman constant. The recorded state of the art is $\Lambda \leq 0.2$: when Platt–Trudgian verified the Riemann hypothesis to the exact height $T = 3,000,175,332,800$ ($\approx 3.0002 \times 10^{12}$), it was noted at once that this improves the Polymath15 bound $\Lambda \leq 0.22$ to $\Lambda \leq 0.2$. A sharper, essentially 0.197-class computation at the same barrier site was publicly reported in a May 2020 comment on T. Tao’s blog by Rudolph [Dwars] — uncertified, unrefereed, and never independently verified. This paper supplies the certified, theorem-grade version of that bound and goes beyond it. Subject to the published Polymath15 criterion and the published Platt–Trudgian verification, the attached certificates prove, unconditionally,

$$\Lambda \leq \frac{3}{16} = 0.1875 \text{ exactly,}$$

the deepest rung of a certified assembly ladder $0.1965 \rightarrow 0.1891 \rightarrow 0.1875$ at the same barrier site, every rung a certified improvement below the recorded $\Lambda \leq 0.2$ and below the uncertified 0.1972624050 sighting of 2020, with every numerical ingredient backed by interval-arithmetic certificates (explicit ball enclosures, exact-rational endpoint comparisons, exit-0 standalone verifiers) and every theorem-grade claim verified on two mechanically independent code lines sharing zero code. The route to the 0.1965 rung runs through a certified Euler-3 mollifier bound, certifiably beyond the reach of the Euler-2-class mollifier underlying the 2020 computation; its assembly binds a certified Euler-3 selection chain, gap-free finite sweeps, a box-uniform tail bound, a three-stage normalization-conversion lemma chain, the y -quantifier transfer, and the site bookkeeping lemmas, with an independent second-line sign-off on the assembled statement. The two deeper rungs are closed by a certified window-tiling assembly route (Section 5): the integer window range is partitioned at certified split points, each piece closed by an independently certified finite-window or tail certificate, and every joint gated in exact rational or outward-rounded interval arithmetic by a standalone assembly verifier. The same certificate chain certifies the intermediate record $\Lambda \leq 0.197$ and the 2020 comment’s exact row, $\Lambda \leq 0.197262405$. Beyond the bound itself we contribute: certified mollifier results, including the first quantitative saturation map of the finite-Euler-product mollifier family at this site (the family provably runs out of cheap depth); the verification methodology, which addresses the cheap-re-verifiability question raised in the 2020 thread and catches a class of silent rendering defects — three of which we document, with resolutions, in the committed Polymath15 artifacts; and a certified negative/feasibility map for the next rung (where the Euler-2 mollifier provably dies, an exact certified costing of the $\Lambda \leq 0.19$ wall at $\geq 3.4881 \times$ the entire Platt–Trudgian computation, a measured GPU pricing of that wall, and a measured kill of the contour-mode verification route). Results currently resting on a single certified verification line are explicitly marked.

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“Mosaic Intelligence” is a collective pseudonym, in the tradition of D. H. J. Polymath. See Section 11 for a full disclosure of how this research was conducted. Contact and updates: [@111111 · github.com/463464q435q43](https://github.com/463464q435q43).

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1. INTRODUCTION

1.1. The de Bruijn–Newman constant. For $t \in \mathbb{R}$ let

$$H_t(z) := \int_0^\infty e^{tu^2} \Phi(u) \cos(zu) du, \quad \Phi(u) := \sum_{n=1}^\infty (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}},$$

so that $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2})$ where ξ is the Riemann xi function. De Bruijn [dB50] showed that the zeros of H_t are all real for $t \geq 1/2$, and Newman [New76] that there is a finite constant Λ — the de Bruijn–Newman constant — such that the zeros of H_t are all real precisely when $t \geq \Lambda$. The Riemann hypothesis is the assertion $\Lambda \leq 0$, and Rodgers–Tao [RT20] proved $\Lambda \geq 0$ (simplified by Dobner [Dob21]), so the lower bound is final modulo RH itself.

On the upper-bound side, the modern record chain is: $\Lambda < 1/2$ [KKL09]; $\Lambda \leq 0.22$ by the Polymath15 project [Pol19]; and $\Lambda \leq 0.2$, obtained when Platt–Trudgian [PT21] verified RH to the exact height $T = 3,000,175,332,800$ ($\approx 3.0002 \times 10^{12}$), which makes the corresponding conditional row of [Pol19] (Table 1) unconditional — as Tao noted in the same blog thread within days of [PT21] appearing [Tao20a]. One week after that note, an uncertified $\Lambda \leq 0.1972624050$ computation at the same barrier site was publicly reported in the same thread [[Dw20]; it was never certified, refereed, or independently verified, and the recorded bound remained $\Lambda \leq 0.2$ (Section 1.2). The honest history of the bound at this RH height is therefore $0.22 \rightarrow 0.2 \rightarrow (\text{uncertified } 0.1972624050) \rightarrow \text{this paper: } \Lambda \leq 0.197$ is a certified improvement below the recorded $\Lambda \leq 0.2$ consequence of [PT21, Tao20a], and $\Lambda \leq 0.1965$ improves further and lies below the earlier uncertified 0.1972624050 sighting [[Dw20] at the same site. The certified window-tiling assembly ladder of Section 5 extends the record to $\Lambda \leq 0.1891$ and then $\Lambda \leq 3/16 = 0.1875$ exactly, at the same site and the same [PT21] height; to our knowledge no prior sighting, certified or otherwise, exists below 0.1965 .

1.2. Prior art: the 2020 blog-comment computation. The headline value of this paper must not be read as a new numerical insight, and we begin by saying so plainly.

On 1 May 2020, in the comment thread of Tao’s Polymath15 announcement post, Rudolph [Dwars] — a Polymath15 contributor — publicly reported an essentially 0.197 -class calculation [[Dw20]. Quoting the Polymath15 paper’s own remark that its Table-1 entries fall between RH-verification heights and that “some extra decimals could be wrought out of the calculation. We have not pursued this”, the comment pursues it: the new height $H = 3000175332800$ of [PT21] translates to $X = 2H$, $N = 690988$; at this N the mollified lower bound “becomes and stays larger than 0.03 for the choice $t_0 = y_0 = 0.1809$ ”; these parameters yield “ $\Lambda \leq 0.1972624050$, and to be on the safe side let’s take 0.1973 ”. The comment further reports an “optimal barrier location” at 6000000185827 and that “running the barrier computations yields a winding number of 0 ”, linking a raw output file in the Polymath15 code repository [Pol18] as evidence.

That comment is, to our knowledge and searches, the strongest prior art on this bound: public, timestamped, parameter-exact — and uncertified. It was never written up, submitted, or refereed; the linked evidence is a raw 2018-toolchain output text file, with no interval enclosures, no stated error radii, and no machine-checkable artifact chain; it received no reply and no independent verification in the thread (the next comment is dated July 2021 and is unrelated). The published record remained $\Lambda \leq 0.2$.

What this paper adds, relative to that comment, is the subject of the next subsection. What it does *not* add is the discovery that 0.197-class values are reachable from the Platt-Trudgian height: that was understood, publicly, on Tao’s own blog, in May 2020. Our certificate chain in fact certifies the comment’s exact row as stated ($\Lambda \leq 0.197262405$ exactly, at $t_0 = y_0 = 0.1809$; Corollary 3.2), and a re-tuned parameter row at the same barrier site sharpens it to $\Lambda \leq 0.197$ exactly.

1.3. What this paper does. The headline claim, stated with its trust base explicit: *subject to the published Polymath15 criterion (Theorem 1.2 of [Pol19]) and the published Platt-Trudgian verification ([PT21], Theorem 1, exact height $T = 3,000,175,332,800$), the attached certificates prove unconditionally that $\Lambda \leq 3/16 = 0.1875$ exactly*, the deepest rung of a certified assembly ladder that also certifies $\Lambda \leq 0.1891$, $\Lambda \leq 0.1965$, the intermediate record $\Lambda \leq 0.197$, and the 2020 comment’s row $\Lambda \leq 0.197262405$. The contributions below describe what those certificates are and how they are verified.

(a) *Certification (Section 3).* The first theorem-grade version of the 0.197-class bound: every hypothesis of the Polymath15 criterion is discharged by machine-checkable artifacts — interval/ball enclosures with stated radii, exact-rational endpoint comparisons, standalone verifiers re-run from clean copies with exit 0 — and every theorem-grade ingredient is verified on *two* mechanically independent code lines (a C/Arb line [Joh17] and a from-scratch Python interval line [J+23] sharing zero code, with formulas independently transcribed from [Pol19]). The thread’s prior art offers one uncertified run and a raw output file; this paper offers a certificate chain that a referee can re-verify mechanically without trusting any part of the producing system.

(b) $\Lambda \leq 0.1965$ (Section 4). Genuinely beyond the 2020 comment: outside the value it reported, and — as we certify — unreachable by the Euler-2-class mollifier its calculation relied on (the certified Euler-2 selection maximum at the 0.1965 row is $0.0201508110 < 0.03$; Section 8). The route runs through a certified Euler-3 mollifier bound ($0.1848895863 > 0.03$ at the same row). Every leg of the 0.1965 assembly is certified and stated in Section 4, including the y -quantifier transfer that was the route’s last open mathematical step. The combined assembly certificate and its independent same-day second-line sign-off (zero shared code, no discrepancy found) close the chain: the two verifiers recompute — rather than re-read — the interval enclosures, the closed-tile coverage of every quantifier domain, the exact rational parameter identities, and the machine-derived floor truncations behind every displayed digit string, comparing each against the frozen manifest (52 and 42 such checks respectively, both exiting 0; Section 4.5).

(b') *The record ladder $\Lambda \leq 0.1891$ and $\Lambda \leq 3/16 = 0.1875$ (Section 5).* The headline: a certified window-tiling assembly route descends the record below 0.1965 in two rungs at the same barrier site. The integer window range is partitioned at certified split points (each covered by both adjacent legs), each piece is closed by an independently certified finite-window or tail certificate, and a standalone assembly verifier gates every joint in exact rational and outward-rounded interval arithmetic (58 checks per assembly, exit 0, re-run twice from clean copies). The 0.1875 assembly’s tail leg is re-derived live inside its own verifier, so the assembly cites no uncertified input; an independent dual second-line sign-off (zero shared code, enforced by an AST audit; 76 gates, exit 0) re-proves that leg on its own engine with strictly sharper endpoints. To our knowledge no prior sighting, certified or otherwise, exists below 0.1965.

(c) *Improved mollifier bounds and family saturation (Section 6)*. Certified bounds for the finite-Euler-product mollifier family (Euler-2/3/5/7) in both published lemma variants, including: the Euler-3 bound that unlocks the 0.1965 row; an improved-triangle-inequality variant under which plain Euler-2 already clears the 0.1965 selection threshold; and a quantitative *ladder saturation* map at the barrier site — certified 0.03-crossover brackets in t_0 for each family member, showing the marginal depth per added prime collapsing ($\approx 0.0094 \rightarrow \approx 0.0025 \rightarrow \approx 0.0010\text{--}0.0015$) while per-point cost grows $\approx 20\times$: the “just add more primes” branch of the design space is closed at this site.

(d) *Verification infrastructure (Section 7)*. In the same 2020 thread, Tao remarked that verifications of this class are “not going to be quickly replicable”, that converting one into a “probabilistically checkable proof ... seems more effort than it is really worth”, but that “checksums ... might be doable without having to run the entire code again” [Tao20b]. The certificate discipline of this paper is a working answer to that musing: every claim maps to a self-contained artifact whose verification costs minutes, not the production cost; spot checks are seeded by the hash of the audited artifact itself (unforgeable, reproducible); all decimal digit strings obey a machine-derived floor-truncation law; and transfers between machines carry manifests and verify scripts. A Lean 4 component accompanying the artifact release machine-checks the implication skeleton of the $\Lambda \leq 0.197$ chain (headline theorem proved sorry-free at pinned toolchain versions from nine named axioms, axiom-footprint and literal-consistency audits passing; the 0.1965 chain is staged as follow-on — Section 7.5). The discipline is not decorative: it caught the three upstream artifact defects of Section 9 and several internal ones.

(e) *A certified negative/feasibility map (Section 8)*. Where the next improvements do and do not live, with certificates: the Euler-2 mollifier provably dies before 0.1965 at this site (grid-sharp); constant-level tuning at fixed RH height is certifiably worth less than 10^{-8} in Λ ; the $\Lambda \leq 0.19$ rung requires RH to $T \geq 10000000065626$ exactly — a certified $\geq 3.4881\times$ the entire Platt–Trudgian computation, upgrading the Polymath15 paper’s “a little higher than 10^{13} ” remark to certified numbers; a measured GPU hint/certify pricing of that wall (no sub-0.7 core-ms/zero certified tier demonstrated; the binding component isolated and priced); and a measured kill of the contour-mode (zero-free strip) verification route, $354.8\times$ below its own pre-registered viability line.

1.4. Verification standard. Every numerical result quoted in this paper is either an exact rational or a floor truncation of a certified interval endpoint (Convention 2.2); comparisons against thresholds are ball-arithmetic certificates (`arb_gt/arb_lt`) or exact rational comparisons of binary interval endpoints — never floating-point reads. Throughout, a *check* (synonymously, a *gate*) is one machine-evaluated pass/fail assertion inside a standalone verifier: it recomputes — never re-reads — a stated fact (an interval enclosure, a closed-tile coverage or containment of a quantifier domain, an exact rational parameter identity, or a machine-derived floor/ceiling truncation compared against the frozen manifest string) and fails the entire verifier on any mismatch; quoted check counts are secondary bookkeeping, never the content of a claim. Where a claim currently rests on a single certified verification line rather than two independent ones, that status is stated explicitly (Table 1). All certificates and checkers are published in the artifact bundle (Section 12), and a front-page theorem map (`THEOREM_MAP.md` in the bundle root, with a machine-readable `THEOREM_MAP.json`) lists, for every certificate leg of both chains, the exact statement it proves, its bundle path, its manifest digests, its toolchain pins, its frozen expected output, and its run command.

1.5. Organization. Section 2 fixes the Polymath15 criterion, the barrier site, and the digit conventions. Section 3 proves the intermediate record $\Lambda \leq 0.197$. Section 4 proves the headline bound $\Lambda \leq 0.1965$. Section 6 contains the mollifier results; Section 7 the verification methodology; Section 8 the negative/feasibility map; Section 9 the upstream-artifact findings; Section 10 states what is *not* claimed; Section 11 discloses how the research was conducted; Section 12 maps every claim to its certificate.

2. PRELIMINARIES

2.1. The Polymath15 criterion. Our engine is Theorem 1.2 of [Pol19], which we restate in the normalization used throughout (the zero $z = x + iy$ of H_t corresponds to the zeta zero $\sigma + iT$ with $x = 2T$, $y = 1 - 2\sigma$; this map and its spot checks are part of the certified package).

Theorem 2.1 ([Pol19, Thm. 1.2]). *Let $X > 0$, $t_0 > 0$, $0 < y_0 \leq 1$ satisfy $y_0^2 + 2t_0 \leq 1$. Suppose that*

- (i) *the zeros of $H_0(x + iy)$ with $0 \leq x \leq X$ are all real (equivalently, RH holds for zeta zeros up to height $T = X/2$);*
- (ii) *$H_t(x + iy) \neq 0$ on the barrier slab $x \in [X, X + 1]$, $y \in [y_0', 1]$, $t \in [0, t_0]$ (verified as a winding number 0 around mesh rectangles), where y_0' is the slab floor; and*
- (iii) *the effective Dirichlet-series lower bound keeps $|f_t| > 0$ above its approximation error for all $x \geq X$ at the relevant heights y .*

Then $\Lambda \leq t_0 + y_0^2/2$.

Hypothesis (iii) is evaluated through the effective approximation $f_t = (A^{\text{eff}} + B^{\text{eff}})/B_0^{\text{eff}}$ of [Pol19, Thm. 1.3], with the explicit error terms e_A, e_B, e_{C_0} of displays (20)–(24)/(71)–(74) there, and the truncation length $N = N(x, t) = \lfloor \sqrt{x/4\pi + t/16} \rfloor$. The mollified lower bounds are the Lemma-10.1/Section-8 family of [Pol19], in two published variants: the *T-type* (naive triangle) and *L-type* (improved triangle inequality) branches, each admitting finite-Euler-product mollifiers $\prod_{p \leq P} (1 - b_p^t p^{-s})$ (“Euler- m ”; the published unconditional runs used $P = \{2, 3, 5, 7\}$). The selection threshold 0.03 is the paper’s own stated target for the mollified bound.

2.2. The barrier site. All computations in this paper live at the barrier site

$$X = 6000000185827,$$

the committed barrier location of the Polymath15 repository [Pol18] and of the 2020 comment [[Dw20]. Hypothesis (i) is supplied at this site *only* by [PT21]: $X/2 = 3000000092913.5 \leq T_{\text{PT}} = 3000175332800$, with exact slack $350479773/2$. The site is within 5.85×10^{-5} (relative) of the ceiling $X = 2T_{\text{PT}}$ that hypothesis (i) imposes at this RH height — it is, in this exact sense, the last barrier site the Platt–Trudgian height can serve.

Two parameter rows at this site appear throughout:

- the *record row*: $t_0 = 71/400 = 0.1775$ exactly, $y_0 = \sqrt{39/1000}$ (floor truncation 0.1974841765), $N_0 = 690988$, yielding $\Lambda \leq t_0 + y_0^2/2 = 197/1000$ exactly;
- the *slab-corner row*: $t_0 = y_0 = 0.1809$ (the 2020 comment’s parameters), yielding $\Lambda \leq 0.197262405$ exactly (a terminating decimal).

The record row’s barrier region $[\sqrt{0.039}, 1] \times [0, 0.1775]$ is strictly contained in the certified slab $[0.1809, 1] \times [0, 0.1809]$ (containment certified in exact rationals), so one slab certification serves both rows.

2.3. Digit conventions.

Convention 2.2 (floor truncation). “Floor-truncated to d digits” means the decimal string of $\lfloor x \cdot 10^d \rfloor / 10^d$, rounding toward $-\infty$ for *every* sign, so that $\text{trunc}(x) \leq x < \text{trunc}(x) + 10^{-d}$. A floor-truncated rendering of a certified lower endpoint is therefore itself a certified lower bound; upper bounds are rendered by the corresponding ceiling and labeled as such. Every digit string in this paper is machine-derived from the exact value inside the artifact (binary mantissa/exponent extraction; no decimal round-trip, no hand transcription); the artifact’s verify script re-certifies every displayed literal. We adopt this convention because the default printing routines of common ball-arithmetic libraries round to *nearest* — a trap responsible for every one of the upstream artifact findings of Section 9 and several internal incidents.

Convention 2.3 (two verification lines). A claim is *theorem-grade* in this paper only when verified on two mechanically independent code lines sharing zero code (different language and library where possible: C/Arb [Joh17] versus pure-Python interval arithmetic [J⁺23] or exact **Fraction** integer arithmetic, with formulas independently transcribed from the [Pol19] TeX source). Single-line results are marked in Table 1 and never stated as theorems.

3. THE CERTIFIED RECORD $\Lambda \leq 0.197$

Theorem 3.1 (main). *Unconditionally,*

$$\Lambda \leq \frac{197}{1000} = 0.197.$$

Corollary 3.2 (the 2020 comment’s row, certified). *Unconditionally, $\Lambda \leq 0.197262405$ (exact terminating decimal), at the parameters $t_0 = y_0 = 0.1809$, $X = 6000000185827$, $N_0 = 690988$ reported in [Dw20].*

The proof is Theorem 2.1 instantiated at the record row (resp. slab-corner row), with each hypothesis discharged by the certified legs below. The sole height input anywhere in the chain is [PT21]. We describe each leg with its two verification lines; the full claim-to-certificate map is Table 5.

3.1. Leg (i): the RH height. $X/2 = 3000000092913.5 \leq T_{\text{PT}} = 3000175332800$ [PT21, Thm. 1], exact slack $350479773/2$; the admissibility conditions $y_0^2 + 2t_0 = 197/500 \leq 1$ and the containments of Section 2.2 are exact-rational checks, re-derived independently on both lines. (We note for Section 8 that hypothesis (i) consumes only a zero-free *strip* of half-width $y_0/2$; a certified strip-sufficiency lemma makes this precise.)

3.2. Leg (ii): barrier winding number 0. *Line 1 (C/Arb).* The Polymath15 production programs (StoredSumsThreaded.c, TloophthreadedV4.c of [Pol18]) were built from source against FLINT/Arb and certified on the published baseline row before use (byte-level artifact agreement plus independent interval spot enclosures). At the campaign site the stored-sums Taylor matrix (62×62 , $N_0 = 690988$, expansion point 345494, 20-digit guarantee) was regenerated and matches the sha256-pinned committed artifact on *all* 7692 data fields by exact string equality. The barrier run covers the *full* slab $x \in [X, X + 1]$, $y \in [0.1809, 1]$, $t \in [0, 0.1809]$: 444 rectangles, every per-rectangle winding field 0.0000000000, overall winding 0, contiguous monotone t -march with no gap, minimum over rectangles of the minimum mesh modulus $|(A+B)/B_0| = 1.4850313937$ (floor-truncated) — five orders of magnitude above the certified approximation error budget (leg (iv)). The run digit-matches the 2018 committed artifact on 444/444 rectangles to ≥ 20 common leading digits, providing an independent

re-validation of the artifact cited by [[Dw20]. Measured cost: 34s wall (255.5 cpu-s) — reproducibility is cheap at this site.

Line 2 (zero shared code). Two independent supplements: (1) a from-scratch pure-Python interval recomputation (160-bit directed real intervals; complex operations, α_1 , H_{01} , the $(A+B)/B_0^{\text{eff}}$ approximant and the full boundary mesh re-derived from the [Pol19] definitions) of two *hash-chosen* rectangles — indices 326 and 152, selected by the sha256 of the Line-1 output itself (Section 7), so the producer cannot anticipate them — certifying at all 6600 resp. 21736 mesh points that the enclosure excludes 0, that consecutive mesh ratios have certified positive real part, and that the interval winding sum encloses 0 within $(-1, 1)$: certified mesh winding 0, with the Line-1 printed minima contained in the Line-2 directed enclosures; and (2) a hash-seeded corner audit (130-bit intervals, independent transcription) of rectangles 1, 70, 195 and 444 (the $t = 0$ base, two hash-drawn interior rectangles, and the t -march boundary), with containment, non-vanishing, and enclosure-width gates.

The mesh-density adequacy step (mesh winding 0 $\Rightarrow$ continuous winding 0) rests on the Line-1 certified derivative bounds, as in [Pol19]; the N -constancy lemma below guarantees the approximant is the *same* Dirichlet polynomial across the slab.

3.3. Leg (iii): the Dirichlet lower bound. *Finite sweep (Line 1).* The Lemma-10.1 Euler-2 mollified selection bound (analytic erfi-tail variant) exceeds 0.03 for *every* integer $N \in [690988, 5000000]$: a gap-free grid of 4,309,013 contiguous points, each value a machine-certified 12-decimal floor truncation with a per- N ball certificate (> 0.03), grid minimum 0.031340098433 at $N_0 = 690988$, maximum 0.519615545572 at $N = 5000000$; plus an analytic tail leg recomputed live by the verifier: 18 doubling-segment interval hulls covering $[5 \times 10^6, 1.31072 \times 10^{12}]$ (every segment certified > 0.03) and a zeta-majorant closed form certifying the bound > 0.03 for all real $N \geq 10^{12}$. Adversarial self-audit: two hash-selected grid values are recomputed live through an independent no-Taylor direct path and a naive full-sum reference.

Finite sweep (Line 2). A zero-shared-code block-uniform certificate (64-bit directed intervals, fixed partition into singletons plus geometric subblocks): for all x with $N(x) \in [690988, 9999999]$ at $y = y_0$, $|f_{t_0}(x + iy_0)| - (e_A + e_B + e_{C_0}) \geq 2.119901028 \times 10^{-2}$ and the selection quantity $\geq 2.745245414 \times 10^{-2}$ (floor-truncated) — a gap-free independent second line over the whole finite range.

Unbounded tail. A mollified tail package: with the Euler-2 mollified per- n triangle bound, $D(N) \leq D(N_1) < 1$ for all $N \geq N_1 = 5 \times 10^6$ and all $y \in [y_0, \sqrt{0.645}]$, with N -monotonicity proved by four certified negative log-derivative gates and a y -monotonicity layer; hence $|f_{t_0}| \geq 0.2477133542$ and $|H_{t_0}/B_{t_0}| \geq 0.2476158451$ (floor-truncated) uniformly on the tail. Line 2 covers $N \geq 10^7$ by an independent per- n absorption derivation, overlapping the Line-2 finite sweep at $[5 \times 10^6, 10^7]$ — jointly gap-free. (A box-uniform generalization covering every row of Section 4 simultaneously is also certified; see there.)

y-quantifier transfer. The sweep certificates above are computed at the single height $y = y_0$. Two certified theorems transfer them to the full hypothesis range $y \in [y_0, \sqrt{1 - 2t_0}]$: a *y-quantifier-reduction* theorem for standard window majorants with y -independent nonnegative coefficients (seven certified interval gates; e.g. $Y_1(N_0) \leq -6.7029 < 0$), and — because the certified sweep functional has *y-dependent* mollifier weights, outside that form — a dedicated *y-monotonicity bridge*: $\text{BOUND}(N, y) \geq \text{BOUND}(N, y_0)$ for every integer $N \in [690988, 5 \times 10^6]$ and every $y \in [y_0, 1]$, certified on an 18-block geometric cover with y treated as a whole interval.

Binding lemma. A record-binding package closes the remaining algebra: (R1) the sweep functional is exactly of standard-window-majorant form (the complex phase cancels identically; verified symbolically), so the reduction theorem applies; (R2) the mollifier conversion $|f_{t_0}| \geq 0.03/C_\lambda \geq 0.0231663603$ (floor-truncated) uniformly for $N \geq N_0$, $y \in [y_0, y_{\max}]$; (R3) a global uniform error budget $e_A + e_B + e_{C_0} \leq 0.000000104590$ (ceiling-truncated) over all windows; and (R4) the single binding inequality

$$m_{\min} - E_{\max} \geq 0.0231662557 > 0 \quad (\text{floor-truncated}),$$

which is hypothesis (iii) with certified margin. Line 2 re-derived R1–R4 from scratch (different majorant chains: p -series integral tail versus endpoint-cap lemma; 35/35 gates, exit 0), with final budgets agreeing — a genuine two-line consistency check, since the intermediate monotonicity chains differ.

3.4. Leg (iv): the error-term audit. The effective-approximation error displays (20)–(24)/(71)–(74) of [Pol19] were implemented in two mechanically independent interval transcriptions (paper-direct head-first summation versus log-space tail-first with an independently derived tail enclosure) and cross-checked at five campaign points; the certified total $e_A + e_B + e_{C_0}$ at the record row is $\leq 1.0466 \times 10^{-7}$ — four orders below the 0.03 selection threshold and five below the slab’s minimum mesh modulus. One transcription deviation in the repository’s error code was found and is reported in Section 9.

3.5. Leg (v): slab bookkeeping. Two lemmas pin the quantifier structure of the assembly: N -constancy — $N(x, t) = 690988$ everywhere on the full marched slab $[X, X + 1] \times [0, 0.1809]$, with robustness margins $\text{frac}(S_{\text{lo}}) \geq 0.3096430277$ and $1 - \text{frac}(S_{\text{hi}}) \geq 0.6903569064$ (floor truncations of exact endpoint rationals); and *window gluing* — uniformly over every $t_0 \in [0, 0.1809]$, the N_0 sweep window covers $[X, X + 1]$ with exact margins ≥ 5377392.8789 and ≥ 11989041.1415 , so no window boundary crosses the slab for *any* admissible row at this site. Line 2 for both is a new toolchain class entirely: pure-integer arithmetic (**Fraction** + **isqrt**, π enclosed from scratch by two intersecting Machin-type brackets), importing no interval library at all.

3.6. Assembly and verification table. The combined certificate manifest binds legs (i)–(v) at the record row, re-runs every bundled verifier live (exit 0), and re-derives the exact assembly $\Lambda \leq t_0 + y_0^2/2 = 197/1000$. The slab-corner instantiation (Corollary 3.2) follows from the same slab and the committed artifact row’s own certified selection bound ($0.0302414398 > 0.03$). An independent record-binding second line signed off on the assembled statement the same day. The full chain was completed on 2026-06-12.

4. THE CERTIFIED BOUND $\Lambda \leq 0.1965$

Theorem 4.1 (headline). *Unconditionally,*

$$\Lambda \leq \frac{1965}{10000} = 0.1965.$$

The proof is Theorem 2.1 instantiated at the stretch row below, with every leg certified and the combined assembly certificate plus its independent second-line sign-off described in Section 4.5.

ingredient	line 1	line 2	lines
height admissibility (i)	exact rationals in-manifest	exact rationals, independent	2
stored-sums matrix	exact match to pinned artifact	100% re-march, 7688/7688	2
winding 0 (ii), 444 rects	C/Arb full slab	hash-chosen rects + corners	2
finite sweep (iii), $[N_0, 5 \cdot 10^6]$	per- N ball certs, gap-free	block-uniform, gap-free	2
tail (iii), $N > 5 \times 10^6$	analytic tail + mollified pkg	per- n absorption, $N \geq 10^7$	2
y -transfer	reduction thm + bridge	independent R1–R4 re-derivation	2
binding $m_{\min} - E_{\max} > 0$	50/50 checks, exit 0	35/35 gates, exit 0	2
error audit (iv)	transcription T1	transcription T2	2
N -constancy / gluing (v)	interval line, prec 250	pure-integer line	2

TABLE 1. Verification-line status of every ingredient of Theorem 3.1. Every row carries two mechanically independent lines with zero shared code. The corresponding table for the $\Lambda \leq 0.1965$ assembly is in Section 4.

4.1. The row, and why it needs a new mollifier. The stretch row is

parameter	value	rendering	status
X (barrier site)	6000000185827	exact integer	as in Section 2.2
t_0	$177/1000 = 0.1770$	exact	certified
y_0	$\sqrt{39/1000}$	0.1974841765 (floor)	certified
N_0	690988	exact integer	certified (N -constancy reused)
$y_{\max}$	$\sqrt{1 - 2t_0} = \sqrt{646/1000}$	exact	certified
Λ bound	$t_0 + y_0^2/2 = 1965/10000$	exact	certified
admissibility	$y_0^2 + 2t_0 = 393/1000 \leq 1$	exact	certified
height	$X/2 = 3000000092913.5 \leq T_{\text{PT}}$	slack 350479773/2 exact	certified

The Euler-2 selection bound that carried Section 3 provably cannot carry this row: over a 19-point t_0 grid the certified Euler-2 maximum at $\Lambda = 0.1965$ is $0.0201508110 < 0.03$ (grid-sharp; ball certificates on both endpoints), a certified deficit factor ≥ 1.488 (Section 8). The route therefore runs through the Euler-3 mollifier ($P = \{2, 3\}$, divisors $\{1, 2, 3, 6\}$).

4.2. Certified legs.

- *Selection anchor (Euler-3).* At the row point ($N = N_0$), the T-type Lemma bound with the Euler-3 mollifier is certified $0.1848895863 > 0.03$ (ball radius $\leq 10^{-26}$), with a like-for-like Euler-2 control $0.0147279901 < 0.03$ at the identical point — the gain is the mollifier shape, not the tail treatment. Cross-checked to 25 digits against the repository’s own pari implementation [The24]; independently reproduced on a second interval line.
- *Finite sweep, full grid.* The Euler-3 T-type bound is certified > 0.03 at *every* integer $N \in [690988, 5000000]$ — 4,309,013 contiguous points, each individually ball-certified, no sampled-grid-versus-continuum gap — at $t = 0.1770$ (min 0.184889586327 at N_0 , max 0.487572439544) and at $t = 0.1775$ (min 0.191442648559 , max 0.490581145542); and — t -box-uniformly — for every real $t \in [0.1770, 0.1775]$ simultaneously, with box-uniform per- N lower bounds (box minimum 0.168072461950), the continuous t -coverage obtained by ball inclusion rather than sampling. These sweeps are computed in the production evaluator’s normalization; the conversion lemmas below carry them to the form the binding theorem consumes.

- *Row-parameterized binding theorem.* The record-binding package of Section 3.3, parameterized in t_0 and extended from Euler-2 to multi-prime Euler mollifiers $\prod_{p \in P}(1 - \beta_p)$, $P \subseteq \{2, 3, 5, 7\}$, instantiated at this row: the multi-prime phase cancellation is verified symbolically (the selection functional is again of standard-window-majorant form with y -independent masses, so the y -quantifier-reduction theorem applies); the conversion constants are certified (σ -floor ≥ 1.7887022679 ; $C_{\{2\}} \leq 1.2956514986$, $C_{\{2,3\}} \leq 1.4871894818$, $C_{\{2,3,5,7\}} \leq 1.6384739261$, ceiling-truncated); the global error budget at the row is $e_A + e_B + e_{C_0} \leq 0.000000106981$ (ceiling-truncated), uniform over all windows $N \geq N_0$ and all $y \in [y_0, y_{\max}]$ and independent of the mollifier choice; and the conditional binding theorem reduces the whole assembly to a single sweep hypothesis: a selection certificate $\geq \theta$ at $y = y_0$ for every integer $N \in [N_0, 5 \times 10^6]$ suffices whenever θ exceeds the analytic floor $C_P \cdot E_{\max}$ — certified ≤ 0.000000159101 for Euler-3, so the 0.03 design threshold clears it by a factor ≥ 188560.6 , and the binding margin at $\theta = 0.03$ is ≥ 0.0201721712 (floor-truncated). Anchor checks: the same parameterized code at the record row reproduces every record-binding digit string of Section 3 at its stated rounding mode.
- *Normalization-conversion chain (three certified stages).* The sweep certificates above are stated in the production evaluator’s normalization (per- N exponent, mollifier-normalized, with a subtracted correction term); the binding theorem consumes a frozen-exponent selection form. A three-stage certified lemma chain closes this seam. *Stage 1 (exponent alignment):* the frozen exponent sits strictly below the production exponent for every $N \geq N_0$, with one-sided gap $< 1.04 \times 10^{-15}$ uniformly in N (proved cancellation-free, not by interval subtraction of the two ≈ 1.789 exponents), so a mass-inflation lemma converts any production-normalization numerator certificate at threshold c to the frozen-form bound at threshold $c - D^*$ with $D^* \leq 1.8 \times 10^{-14}$ — twelve orders below the 0.03 design threshold; box-uniform conversion margin ≥ 0.0124722278 (floor-truncated) over the whole sweep range. *Stage 2 (normalization and leak accounting):* the production evaluator’s subtracted correction term is certified nonnegative on its entire domain (so certificate face values are conservative), its mollifier normalization factor is exactly the binding theorem’s conversion-constant shape (a symbolic polynomial identity, all real exponents at once), and the numerator-recovery lemma yields: every production certificate $\geq 3/100$ on the sweep range gives a numerator bound ≥ 0.0424722278 (floor-truncated), and ≥ 0.2617557548 at the certified grid minimum 0.184889586327. *Stage 3 (y-transfer routing):* a kernel-order lemma (sharp termwise comparison constant certified in $[0.8681942568, 0.8681942569]$ at the row) shows the improved-triangle-inequality grid certificates of Section 6 *cannot* be converted termwise at face value (a certified negative: conversion is possible only above a threshold in $[0.0945520619, 0.0945520620]$, $3.1517 \times$ the design threshold); the assembly instead routes the y -quantifier through the reduction theorem directly: the independent block-uniform certificate over the closed t -box $[0.1770, 0.1775]$ (margin $\geq 6.066621000 \times 10^{-3}$ at $y = y_0$, zero shared code with the grid line) is of standard-window-majorant form by the multi-prime rewrite lemma, the transfer gates are re-certified at $t_0 = 0.1770$ ($Y_1(N_0)$ enclosed two-sidedly in $[-6.7029388683, -6.7029388682]$), and the composite y -uniform floor is $m - E_{\max} \geq 0.0060665140$ (floor-truncated) for all $N \in [690988, 5 \times 10^6]$, all x in the certified windows, and all $y \in [y_0, y_{\max}]$. The y -quantifier transfer — previously the one open mathematical step of this route — is thereby certified.

- *Unbounded tail, box-uniform.* For every $t \in [0.176, 0.181]$, every $y \in [0.1809, \sqrt{0.648}]$ (a range containing the full hypothesis-(ii) interval of every row in play), and all $N \geq 5 \times 10^6$: $|f_t| \geq 0.0834187152$ and $|H_t/B_t| \geq 0.0833079015$ (floor-truncated), with box-uniform monotonicity gates; anchor-validated against the two per-row tail certificates in different regimes. An Euler-3 mollified box tail over the wider box $t \in [0.162, 0.181]$, $y \in [0.1809, \sqrt{0.676}]$ is separately certified ($|H_t/B_t| \geq 0.0062457711$, floor-truncated, uniformly), as is a sharper per-row tail at the stretch row ($|f_{t_0}| \geq 0.2364630457$).
- *Site bookkeeping.* The site-uniform window-gluing lemma of Section 3.5 covers every $t_0 \leq 0.1809$ at this site, hence this row, with the row’s own sharper margins (≥ 5377393.0179 , ≥ 11989041.1446) instantiated and the exact identities $t_0 + y_0^2/2 = 1965/10000$, $y_0^2 + 2t_0 = 393/1000 \leq 1$ certified; two lines as before.
- *Winding and height.* The certified slab of Section 3.2 already contains the row’s barrier region ($39/1000 \geq 0.1809^2$ and $0.1770 \leq 0.1809$, exact), and hypothesis (i) is the same [PT21] input — no new winding or height work is needed.

4.3. The convention seams, tabulated. The three-stage conversion chain above exists because different certificate families state their bounds in different normalizations (“conventions”). Table 2 lists each seam with its certified transfer route. One point deserves explicit emphasis: Stage 3 does *not* use termwise positivity of the improved-triangle-inequality kernel — termwise conversion is certified *impossible* at the certificates’ face values, possible only above a threshold $L^* \in [0.0945520619, 0.0945520620]$ (Section 4.2, stage 3) — and instead routes through the crude-kernel box certificates and the y -quantifier-reduction monotonicity theorem over the full closed domain $y \in [y_0, y_{\max}]$, $N \in [690988, 5 \times 10^6]$.

source convention	target convention	certified transfer route
Stage 1: production evaluator’s per- N exponent	frozen-exponent selection form	one-sided exponent gap ($< 1.04 \times 10^{-15}$ uniformly in N , proved cancellation-free) + mass-inflation lemma; conversion cost $D^* \leq 1.8 \times 10^{-14}$
Stage 2: mollifier-normalized numerator with subtracted correction term	binding theorem’s numerator / conversion-constant form	correction term certified nonnegative on its whole domain + symbolic normalization identity + numerator-recovery lemma
Stage 3: single-height ($y = y_0$) sweep certificates	full closed range $y \in [y_0, y_{\max}]$, $N \in [690988, 5 \times 10^6]$	<i>not</i> termwise positivity (certified impossible below L^*); crude-kernel box certificate over the closed t -box $[0.1770, 0.1775]$ + the y -quantifier-reduction monotonicity theorem

TABLE 2. The convention seams of the 0.1965 chain and their certified transfer routes (conversion stages 1–3 of Section 4.2).

4.4. Verification lines. Every leg above carries its own standalone exit-0 verifier, and the two-line discipline of Convention 2.3 is applied as follows. The row-parameterized binding theorem is re-derived on a second, zero-shared-code line (different freeze family, different tail majorant; 58 gates) whose independently computed constants agree digit-exactly on σ -floor, all three C_P , the error budget, and the binding margin — with the deliberate crudeness of its own tail majorant isolated and bounded ($\leq 3.42 \times 10^{-12}$), so the headroom conclusion is a

two-line fact. Conversion stages 1 and 2 each carry an independent second line (precision 320 versus 120; symbolic sign atoms re-derived; the stage-1 second line additionally resolves the 10^{-15} -scale exponent gap by *direct* two-sided subtraction, a consistency check structurally unavailable to the first line); the conversion constant 1.4871894818 is now reproduced on three mechanically independent lines. The box tail certificates carry zero-shared-code second lines (independent polynomial engine for the coefficient-magnitude lemma; independent floors within 2×10^{-4} , orderings gated). The finite-range zero-exclusion is covered by two independent certificate families: the Euler-3 production-normalization grid (one line, plus adversarial hash-seeded screens) and the block-uniform t -box certificate (zero shared code) that the stage-3 routing theorem consumes. The stage-3 routing theorem also carries an independent second line (zero shared code; 31 gates, exit 0): the kernel-order lemma’s universal quantifier is re-derived by an exhaustive symbolic sign case-split, the conversion-threshold identity and its monotonicity are re-proved symbolically, and the stage-3 digit strings quoted above are machine-reproduced — so every conversion stage now carries two lines.

4.5. The assembly certificate. The 0.1965 analogue of the record package of Section 3 is certified. The combined assembly manifest, shipped in the bundle at `certified1965/assembly_1965/`, is a citation-arithmetic package: every hypothesis of Theorem 2.1 is discharged by a named certified leg of Section 4.2, and the standalone verifier re-derives the exact assembly $\Lambda \leq t_0 + y_0^2/2 = 177/1000 + 39/2000 = 393/2000 = 0.1965$ exactly and gates every joint — the slab containments, the construction identities, the gap-free window tiling of $[X, \infty)$ (finite leg $N \in [690988, 5 \times 10^6]$ spliced to the box tail with certified overlap ≥ 125663718.7099), and the margins — in exact rational and outward-rounded interval arithmetic (52 checks in the sense of Section 1.4, exit 0, re-run from a clean copy). The composite finite-leg floor is re-gated exactly (≥ 0.0060665140 , floor-truncated), and the sole height input in the entire chain is hypothesis (i) with exact slack $350479773/2$. An independent second-line sign-off — bundle directory `certified1965/assembly_secondline/`; zero shared code, 42 checks, exit 0 — re-establishes the exact record arithmetic and every joint on its own line, machine-reproduces the certified anchor values in two distinct parameter regimes, and adds a falsification probe at an in-window point; no discrepancy was found. The chain was completed on 2026-06-12, with the sign-off the same day.

5. THE CERTIFIED RECORD LADDER: $\Lambda \leq 0.1891$ AND $\Lambda \leq 3/16$

The two rungs below 0.1965 are closed by a different route than the single-row sweeps of Sections 3–4: a *window-tiling assembly*. The hypothesis-(ii) region of Theorem 2.1 is partitioned along the integer window parameter N at certified split points, each piece is closed by an independently certified finite-window or tail certificate (the pieces may, and do, use different mollifier classes: zero-freeness of each piece is a mollifier-independent conclusion, so the pieces compose), and a standalone assembly verifier gates every joint — containments, the exact tiling identity, split-point overlap by both adjacent legs, margins, and the record arithmetic — in exact rational and outward-rounded interval arithmetic. Both rungs sit at the barrier site of Section 2.2, at $y_0 = \sqrt{39/1000}$, with hypothesis (i) discharged by [PT21] alone (exact slack $350479773/2$, the sole height input, as everywhere in this paper), and hypothesis (iii) by a certified boundary-winding-0 certificate on the slab $[X, X+1] \times [33/200, 1]$ uniform over all real $t \in [0, 0.1809]$ (independent corner-audit second line) combined with certified N -constancy on the slab, whose two exactness anchors (≥ 5377392.8789 , ≥ 11989041.1415 , floor-truncated) are re-derived inside each assembly verifier at 220-bit interval precision. The

0.1965 leg	status	lines
Euler-3 selection anchor at N_0	certified ($0.1848895863 > 0.03$)	2
Euler-3 full-grid sweeps, $t = 0.1770 / 0.1775$	certified, gap-free	1
t -box-uniform sweep $[0.1770, 0.1775]$ (Euler-3)	certified, gap-free	1
block-uniform t -box certificate (zero shared code)	certified, gap-free	1
binding theorem at the row (multi-prime)	certified	2
conversion stage 1 (exponent alignment)	certified	2
conversion stage 2 (normalization/leak)	certified	2
conversion stage 3 (y -transfer routing)	certified	2
box-uniform tail $N > 5 \times 10^6$ (Euler-2 and Euler-3)	certified + anchor-validated	2
window gluing / N -constancy at the row	certified	2
winding slab containment, height	certified (record slab reused)	2
combined manifest + sign-off	certified (52 checks + 42 gates, exit 0)	2

TABLE 3. Status of the $\Lambda \leq 0.1965$ assembly. “1” rows have one certified line per Convention 2.3; the two sweep families (rows 2–4) are mutually independent certificates of the finite range with zero shared code.

slab-to-hypothesis binding is the one accepted by the 0.1965 assembly, consumed for strictly smaller t -ranges.

5.1. The rung $\Lambda \leq 0.1891$. At $t_0 = 1696/10^4$: $\Lambda \leq t_0 + y_0^2/2 = 1891/10^4 = 0.1891$ exactly. The window range $[690988, \infty)$ is tiled in three legs meeting at $N_{\text{mid}} = 2745000$ and $N_1 = 5 \times 10^6$ (each split point covered by both adjacent legs): a certified mollified finite-window certificate on $N \in [690988, 2745000]$ (closed t -hull containing t_0 , full y -range to $\sqrt{1-2t}$, composite floor ≥ 0.0110533245); a certified middle-window certificate on $N \in [2745000, 5 \times 10^6]$ (t -hull with t_0 at its closed left endpoint, membership exact) composed with a certified y -extension to the full hypothesis ceiling $y_{\text{max}}^2 = 6608/10^4$ (composite y -uniform floor $2669533163/10^{13} > 0$, re-gated in exact rationals); and a certified tail certificate for $N \geq 5 \times 10^6$ (band containing t_0 , full y -range by a certified termwise monotonicity device, final slack ≥ 0.0279753143). The assembly package (bundle directory `certified1875/assembly_1891/`) is pure citation arithmetic over these certified legs; its standalone verifier runs 58 checks, exit 0, re-run twice from a clean copy.

5.2. The record rung $\Lambda \leq 3/16 = 0.1875$. At $t_0 = 1680/10^4$: $\Lambda \leq t_0 + y_0^2/2 = 1875/10^4 = 3/16$ exactly — a terminating decimal whose floor truncation is its exact value. The tiling here needs only one split, at $N_{\text{mid}} = 2745000$, covered by both legs: a certified mollified finite-window certificate on $N \in [690988, 2745000]$ (the closed tile $t \in [1680/10^4, 1685/10^4]$ containing t_0 as its left endpoint, membership exact; block bottom equal to N_0 , so no uncovered sub-block; composite floor ≥ 0.0134320455 , re-gated), and a certified tail certificate for all $N \geq 2745000$ at cutoff $N'_1 = 2745000$ on the band $t \in [1680/10^4, 1696/10^4]$, full y -range, *re-derived live inside the assembly verifier*: box enclosure on the y -hull at head depth $M = 50000$, termwise y -monotonicity above the hull, all 28 monotonicity constants certified negative (binding value -0.033771 , ceiling-truncated), and the cap side conditions re-gated at the cutoff ($D \leq 0.9854851520 < 1$; tail slack $\geq 0.0091331201 > 0$), with the machinery anchored bit-identically against its certified reference regime ($N_1 = 5 \times 10^6$, $M = 20000$):

$D \leq 0.7334562057$, $\text{flow} \geq 0.1747906321$). The live leg means the assembly cites no un-certified input: everything it consumes is either a certified artifact of this bundle or re-derived by its own verifier in the same run. The package (bundle directory `certified1875/assembly_1875/`) runs 58 checks, exit 0, in ≈ 21 s on one core, re-run twice from a clean copy.

Both assembly packages were additionally re-run, exit 0, on a second machine (macOS, system Python) from a byte-identical copy before deposit. Per the marking convention of Section 1.4, the leg certificates carry the verification lines stated above; the independent second-line sign-off on each *assembled statement* is recorded in the bundle at `certified1875/assembly_secondline/`.

6. MOLLIFIER BOUNDS AND EULER-FAMILY SATURATION

All bounds in this section are pure Dirichlet-polynomial interval statements with no RH dependency; rows are admissible at the site of Section 2.2 via [PT21]. “Certified” means an `arb_gt/arb_lt` ball certificate at stated radius, digits floor-truncated per Convention 2.2.

6.1. The family map at the 0.1965 row. At $(X, N_0, t_0 = 0.1770, y_0 = \sqrt{0.039})$, T-type branch:

mollifier	certified bound	verdict vs 0.03
none	in $[-0.7390406377, -0.7390406376]$	fails (the mollifier does all the work)
Euler-2 $\{1, 2\}$	0.0147279901	certified < 0.03
Euler-3 $\{1, 2, 3, 6\}$	0.1848895863	certified > 0.03
Euler-5 $\{1, \dots, 30\}$	0.2300598134	certified > 0.03

The certified Euler-5 marginal gain over Euler-3 is 0.0451702271 (floor-truncated exact difference of ball midpoints, radii $\leq 5 \times 10^{-33}$) at $\approx 2.5\times$ measured per-point cost — the family’s entire useful headroom at this row. Admissibility of every member is by instantiation of the published lemma family (the unconditional Polymath15 runs used $P = \{2, 3, 5, 7\}$); a box-uniform admissibility-cap certificate additionally certifies modulus and phase caps for E2/E3/E5/E7 over the whole campaign box and shows the 0.03 threshold is *not* family-neutral (the zero-freeness floor $0.03/|E|_{\max}$ certifiably decreases with family depth), so fair cross-family comparison must normalize by $|E|_{\max}$.

6.2. The L-type (improved triangle inequality) variant. Under the published L-type branch of the same lemma family (pointwise dominance $\max(|b - a|, c|b + a|) \leq |b| + |a|$ proved, so L-type $\geq$ T-type for every mollifier at every point), plain Euler-2 *already* clears the stretch threshold at the 0.1965 row: certified 0.1070275323 > 0.03 at the exact point where the analytic-erfi Euler-2 baseline certifies $0.0201508110 < 0.03$ and the T-type Euler-2 $0.0147279901 < 0.03$. L-type Euler-3 at N_0 certifies 0.2423994341. Head-grid coverage at $N \in \{690988, \dots, 710000\}$ shows the Euler-2-failing head of the sweep needs no heavier mollifier under this functional (grid-point certificates; no N -monotonicity claimed). A certified kernel-order negative (Section 4.2, stage 3) shows these L-type values *cannot* be converted termwise into the crude-kernel form the binding theorem consumes at face value; they enter the 0.1965 chain as selection-leg evidence under their own quantifiers only.

6.3. Depth brackets and ladder saturation. At the slab floor $y = 1809/10000$ (exact), $N = N_0$, the 0.03-crossover of each family member was bracketed in the t_0 direction by certified fail/pass endpoint pairs (each “bracket” is the conjunction of an `arb_lt` and an `arb_gt` certificate; grid-point statements only, no t -monotonicity claimed). All four rungs

use one byte-identical independent C/Arb source (sha256-pinned), each with a fresh 25-digit pari/gp cross-check:

mollifier	crossover bracket in t_0	Λ -class fails / passes	cost (s/pt, measured)
Euler-2	(0.1743, 0.1744)	0.190662405 / 0.190762405	5.71
Euler-3	(0.1640, 0.1650)	0.180362405 / 0.181362405	10.97
Euler-5	(0.1615, 0.1620)	0.177862405 / 0.178362405	28.59
Euler-7	(0.1600, 0.1610)	0.176362405 / 0.177362405	114.65

The structural reading (*ladder saturation*): the marginal t_0 depth per added prime collapses — $E2 \rightarrow E3 \approx 0.0094$, $E3 \rightarrow E5 \approx 0.0025$, $E5 \rightarrow E7 \approx 0.0010$ – 0.0015 — while measured per-point cost grows $\approx 20\times$ across the table (cost figures are wall-time estimates, not certified quantities). The just-add-more-primes branch of the mollifier design space is closed for post-0.176-class ambitions at this site: deeper rows need larger y , larger N , structurally different coefficients (non-multiplicative optimization), or a different functional — not longer Euler products. Sample-bracket certificates at conditional deeper sites (Section 8) are consistent with this map.

7. THE VERIFICATION INFRASTRUCTURE

This section describes the discipline; it is the substance of contribution (d) and the working answer to the re-verifiability question raised in the 2020 thread [Tao20b].

7.1. Certificates, not screens. Every claim maps to a self-contained artifact directory: writeup, all inputs, and a standalone verify script that re-derives — not re-reads — every certified statement and exits 0 iff all gates pass. Self-containment is enforced mechanically: each bundle is re-run from a clean copy in a scratch location before acceptance. Raw output files (the form of evidence attached to the 2020 comment) are treated as *screens*, never certificates: they enter the chain only after independent recomputation with stated enclosures.

7.2. Two independent lines, adversarial selection. Convention 2.3 (two lines, zero shared code) is applied to every theorem-grade ingredient. Where full duplication is wasteful, spot-checking is made unforgeable: the entries, rectangles, or grid points to be re-verified are selected by seeding a PRNG with the *sha256 of the audited artifact itself*, so the producer cannot anticipate the audited subset and any reader can reproduce the selection. Hypothesis-class facts that other certificates cite carry an elevated bar (an explicit quantifier audit restating the exact $\forall/\exists$ structure certified — box, variables, endpoint openness, monotonicity directions — plus anchor validation: the machinery must exactly reproduce previously-certified values in at least two different regimes before its new claims are citable).

7.3. The digit law. Convention 2.2 is enforced by machine: every displayed digit string, including those inside prose, is re-derived by the bundle’s verify script from exact binary endpoints, with floor/ceiling direction matched to the bound’s direction. This convention exists because the default print path of standard ball libraries rounds to nearest — a silent trap that produced every finding of Section 9 and was caught internally several times by exactly this gate.

7.4. Cheap re-verification, measured. The point of Tao’s “checksums” remark is cost asymmetry, so we report measured re-verification costs at this site: the full stored-sums matrix re-march (all 7688 components, independent line) costs ≈ 45 minutes single-core, or

81.6 s with a certified parallel harness (10.1–11.2 $\times$ measured); the full 444-rectangle barrier re-run costs 255.5 cpu-s; the assembled record manifest re-verifies in minutes; individual claim bundles verify in 1–120 s. Certified GPU kernels (ball arithmetic with directed-rounding intrinsics and exact error-free transforms; containment-checked against CPU enclosures by exact rational endpoint comparison) bring the stored-sums production step to seconds and certified 2×10^9 -point mesh evaluations to under a minute — the re-verification economics of every future rung improve accordingly.

7.5. A machine-checked Lean implication skeleton. A Lean 4 project accompanying the artifact release machine-checks the *implication skeleton* of the record: the assembly logic (numerical certificates as hypotheses $\Rightarrow$ hypotheses of Theorem 2.1 $\Rightarrow \Lambda \leq t_0 + y_0^2/2$ instantiated) and the box-monotonicity lemmas under the hypothesis-class certificates. Its state, reported exactly: the headline theorem $\Lambda \leq 197/1000$ is proved sorry-free at pinned toolchain versions from *nine named axioms* (one per external input or certificate class — the published RH height, the winding slab, the sweep, the tail, the y -bridge, the conversion, the error budget, the bookkeeping lemmas, and the criterion itself), with a machine-run axiom-footprint audit (the theorem’s kernel footprint matches the declared axiom set exactly) and a literal-consistency check tying every Lean literal to the certificate record, both passing from a clean copy; the project was adversarially audited before release. Two caveats are load-bearing and must be read together. First, the scope is *implications only*: this is *not* a formal verification of the record — nothing in it establishes the truth of the numerical certificates (that is the certificate chain’s job), and nothing formalizes H_t /zeta machinery; the certificates enter as named axioms, never as formalized computations. Second, the formal-skeleton coverage applies to the $\Lambda \leq 0.197$ chain *only*; the extension to the 0.1965 assembly is staged as follow-on work, and the record bound $\Lambda \leq 0.1965$ must not be read as carrying equal formal coverage. Within those bounds, this is the form of assurance the formalization community already values, and it composes with (rather than replaces) the certificate chain.

8. CERTIFIED NEGATIVES AND THE FEASIBILITY MAP

Negative results below are certificates, not impressions; each carries the same artifact discipline, and each is cited so that future work starts from the map instead of re-probing.

8.1. No constant-level path below 0.197 at this height. At the fixed RH height T_{PT} , a certified improvement study closes the constant-tuning routes: the conversion $\Lambda \leq t_0 + y_0^2/2$ is exact (no inequality slack to sharpen); the site is within 5.85×10^{-5} relative of the $X = 2T_{PT}$ ceiling; the certified Euler-2 selection bounds give $0.0313400984 > 0.03$ at $\Lambda = 0.197$ but a grid-sharp maximum $0.0201508110 < 0.03$ at $\Lambda = 0.1965$ (deficit factor ≥ 1.488); and the entire certified error budget is worth less than 10^{-8} in Λ against the certified bound-vs- Λ slope ≥ 22.37 . Escape routes are exactly: better mollifiers (Section 6 — this is the 0.1965 route), more height (below), or a different verification modality for hypothesis (i) (also below).

8.2. The 0.19 wall, certified. The published 0.19 row requires RH to $T \geq 10000000065626$ exactly (certified ratio to T_{PT} in (3.33313, 3.33314)) — upgrading the Polymath15 paper’s “H a little higher than 10^{13} ” remark [?, quoted in] Rudolph2020 to certified numbers. Intersecting two independent published zero-counting bounds [BW25, HSW22] gives width-23 integer enclosures of the zero-counting function at the exact heights (cross-validated: the same computation at T_{PT} contains the exact published count 12363153437138 of [PT21]); the derived workload is [30761039153403, 30761039153426] *new* zeros — a certified $\geq 3.4881\times$ the entire Platt–Trudgian computation. At measured stock-engine prices (FLINT Platt-style batched

verification, measured at both the current and target heights with certified amortization envelopes) the purchase is a \$2.6M–\$12M-class decision at single-thread ceilings, with a production floor band of \$0.875M–\$1.75M at literature core-hour anchors — a funding-scale undertaking, not claimed here; the map prices the decision.

8.3. GPU pricing of the wall (measured). A dedicated compute study tested whether current GPU methods reprice the wall, on a fully certified Turing-complete block of 10,272 zeros at $t \approx 3.0002 \times 10^{12}$ (ball enclosures end-to-end, exact integer zero counts, completeness by count equality; independent checker exit 0). Findings: a GPU *hint* tier is cheap (measured $35.986 \mu\text{s}$ GPU per zero; hints validated 1500/1500 against an oracle) and the hint/certify architecture cuts the blind pipeline $4.60\times$ on the same core — but the *certify* tier measured 8.141306 core-ms/zero (exact rational, EPYC pilot core; 3.233527 on a current-generation core, one repetition), and no sub-0.7 core-ms/zero certified tier was demonstrated. The binding component is isolated and priced: one per-grid-point generic-complex-ball table build inside the library (99.7% of certify cost). Conclusion: the wall holds at current art — $\approx$ parity with the [PT21] production rate on modern cores ($\approx$ \$0.66–0.69M flat), $\approx$ \$150–300K *only* conditional on one named, unbuilt double-double port of that component (projection bracket 0.46–0.78 straddles the 0.7 line). Anyone re-pricing the wall should start from this certified block, not from scratch.

8.4. The strip route: killed by measurement, kept by theory. Hypothesis (i) consumes only a zero-free *strip*: a certified lemma shows that a strip certificate $S(w, h)$ ($\zeta(\sigma + iT) \neq 0$ for $1/2 + w \leq \sigma \leq 1$, $0 \leq T \leq h$) with $w \leq y_0/2$, $h \geq X/2$ supplies hypothesis (i), sharp in w ; one strip at $w^* = 0.070710$, $h^* = 30000000061687.5$ would serve seven rows at once (the record row, the 0.1965 row, the slab corner, and the published 0.21/0.20/0.19/0.18 rows — the last two with exact sharpness). The *contour-mode* implementation of that idea is, however, measured dead: at pilot height $\approx 3.0002 \times 10^{12}$, contour-mode certification costs $272.99\times$ more than sign-change-mode per unit height ($354.8\times$ below the pre-registered $1.3\times$ viability line; mechanism isolated: a $720.9\times$ derivative-enclosure wrap factor over segment boxes), and a measured bulk-Platt price makes the kill conservative by a further $\geq 158.35\times$. The route’s tombstone carries explicit reopening conditions (an engine-license event re-opens cheaper sign-change purchasing; new contour mathematics must beat a certified $33254.1\times$ bar).

8.5. Conditional rungs (parameters, not claims). For completeness of the map, certified *conditional* ladder rows at deeper sites: $X = 2 \times 10^{13}$ ($t_0 = 0.169$, $y_0 = \sqrt{0.037}$, selection bound $0.0408064516 > 0.03$) gives $\Lambda \leq 0.1875 = 3/16$ exactly *conditional on RH to $T = 10^{13}$* ; and $X = 6 \times 10^{13}$ ($t_0 = 0.162$, $y_0 = \sqrt{0.034}$, bound 0.0354035885) gives $\Lambda \leq 0.179$ exactly *conditional on RH to $T = 3 \times 10^{13}$* . These are parameter rows with certified selection legs and explicitly stated unmet height obligations — the exchange table for any future height purchase.

9. FINDINGS IN THE COMMITTED POLYMATH15 ARTIFACTS

In the course of certifying against the committed 2018–2021 artifacts of [Pol18], the verification layer surfaced three presentation-level defects. We state them with care: *none affects any published theorem, any Table-1 bound, or any certified chain in this paper* — each is a rendering or labeling artifact of exactly the class our digit law (Convention 2.2) exists to catch, and we report each with its resolution and, where applicable, certified replacement

data. We record them in the spirit of strengthening, not criticizing, a computational record that our own work builds on and re-validates.

Finding 9.1 (Table-1 “lower bound” column provenance). The “lower bound” column entries 0.0341 (the 0.21 row) and 0.0376 (the 0.20 row) of [Pol19, Table 1] are 4-d.p. nearest-rounded renderings of the Lemma-10.1 Euler-2 mollified selection bound on $|1 - \beta||f_{t_0}|$ — certified values 0.0341170740 and 0.0376124001 (floor-truncated, interval radii $\leq 6 \times 10^{-26}$, cross-checked to 25 digits against the repository’s own pari source) — i.e. a bound on the *mollified* quantity, not on $|f_{t_0}|$ itself, and not any output mode of the repository’s `LargexBoundutility`. *Resolution*: a labeling clarification only; the criterion consumes exactly this quantity, so all rows stand. We suggest the column be labeled as the mollified selection bound.

Finding 9.2 (0.21-row winding artifact internal label). The committed winding artifact for the 0.21 row (`windnum_nolemma_x2000000129093_y_0.15492_1_t_0_0.198.txt`) prints $y_0 = 0.15429$ in its own header while its filename and [Pol19, Table 1] give $y_0 = 0.15492$ — a $9 \leftrightarrow 2$ digit transposition of magnitude exactly 6.3×10^{-4} . (Separately, Table 1’s 0.15492 is $\sqrt{2(0.21 - 0.198)} = 0.1549193\dots$ rounded to nearest at 5 decimals; floor truncation would give 0.15491.) *Resolution*: benign in the favorable direction — the header’s smaller y_0 makes the certified region strictly *contain* the intended one, and all 386 rectangles report winding 0, so under either reading the artifact certifies $\Lambda \leq 0.198 + 0.15492^2/2 = 0.2100001032$ exactly; the 0.21 row is unaffected.

Finding 9.3 (0.19/0.18-row stored-sums artifacts not floor-truncation-clean). The committed stored-sums artifacts at the 0.19 row (64×64 , $X = 20000000131252$) and the 0.18 row (67×67 , $X = 60000000123375$) are not clean under a 20-digit floor-truncation reading: 8 (resp. 2) entries — all imaginary components — print floor+1, i.e. nearest-rounded, consistent with the default nearest-rounding print path of the producing library. Each deviation is *proven* from exact interval endpoints by an exact-rational checker run in a classified mode in which any unproven deviation fails hard; two independent computations (a fresh CPU ball-arithmetic run at precision 126 and a certified GPU triple-double kernel) agree on the corrected digits. Additionally, the precision-126 enclosures *straddle* the 20-digit boundary on 5 (resp. 2) further entries — the legacy precision has nearly exhausted its headroom at these deeper rows (worst relative radius $2^{-68.7} \approx 20.7$ digits). *Resolution*: no consequence for any published bound (the artifacts are inputs to *future* rungs, not to any published row’s chain; the 0.20/campaign-row artifacts pass the strict digit law 7688/7688); we recommend regenerating the two artifacts at higher precision under a floor-truncation protocol before the 0.19/0.18 rungs consume them, and our certified dumps are offered as replacements.

One further micro-finding from the error-term audit (Section 3.4): the repository transcription `pari/error_bounds.txt` uses $6.92/(x_N - 6.66)$ where [Pol19, Prop. 6.6(vi)] displays $6.92/(x - 12)$; certified strict (the repository bound is the smaller, i.e. less conservative) with relative magnitude $< 2 \times 10^{-12}$ at campaign x — harmless numerically, noted because our chain cites the paper’s display, not the repository code.

10. LIMITATIONS: WHAT IS NOT CLAIMED

- *No claim of numerical novelty for 0.197-class values.* The reachability of $\Lambda \leq 0.1973$ -class bounds from the [PT21] height was publicly demonstrated in 2020 [[Dw20]; our contribution there is certification (Section 1.2). The re-tuned 0.197 row is a (modest) sharpening within the same site.

- *Saturation statements are grid-point certificates.* The brackets of Section 6.3 are fail/pass certificate pairs at named t_0 grid points; no t -monotonicity is claimed, and the closure statement is scoped to the finite-Euler-product family at the named site, slab floor, and functional family.
- *Cost and dollar figures are sizing arithmetic.* Measured timings are labeled; dollar figures derive from measured anchors and stated rates and are never certified quantities.
- *Single-line results.* Rows marked “1” in Table 3 (the Euler-3 full-grid and t -box sweeps and the block-uniform t -box certificate) currently rest on one certified line each; per Convention 2.3 they are stated as certified computations, not theorems, until their second lines land. (The finite range as a whole is nonetheless covered by two mutually independent certificate families with zero shared code, and the assembled statement itself carries two lines.) Everything in Table 1 carries two lines.
- *The conditional rungs of Section 8 are parameters, never bounds;* each is stated with its unmet height obligation.
- *The GPU study’s projections are conditional.* The \$150–300K path requires an unbuilt component; the measured certify-tier numbers are the demonstrated state of the art.

11. AI DISCLOSURE

This paper reports a fully AI-derived body of results. The campaign design, parameter optimization, interval implementations, certificates, verification tooling, negative results, and the text of this paper were produced by Mosaic Intelligence’s AI research system. Human involvement was strategic direction: target selection, resource approval, and release decisions. All results stated as certified are machine-verified from explicit certificates; the certificates and the independent checkers are published in the artifact bundle (Section 12), so every claim can be re-verified without trusting any part of the producing system. This disclosure follows the practice of our earlier submissions to the `teorth/optimizationproblems` repository, which carried the statement: “This is a fully AI-derived result: the construction was found and certified by Mosaic Intelligence’s automated search-and-verification system, and the submission text was AI-prepared. All numerical results and references were independently re-run and verified before submission.”

12. ARTIFACTS

The results of this paper are accompanied by a self-contained artifact bundle, archived as `dbn21a-certificates` (a `.zip` of the `bundle_21a/` tree described below) and deposited at DOI 10.5281/zenodo.21175533 (the deposit’s identifier; resolves on publication of the deposit). The bundle’s root-of-trust is its `MANIFEST.sha256`, which pins every file by sha256; the front-page entry points carry the digests below.

The first object a reviewer should open is the bundle’s front-page theorem map — `THEOREM_MAP.md` at the bundle root, with a machine-readable `THEOREM_MAP.json` — which gives, one row per certificate leg of both chains: the exact statement the certificate proves, its bundle path, the sha256 manifest digests of its writeup and verifier, its toolchain pins, its frozen expected output lines, and its run command.

Every certified leg of the two bound chains maps to a self-contained directory with a standalone verifier (Python 3 standard library + `mpmath`, or C sources compiled against FLINT/Arb by the bundled script); each verifier exits 0 iff all its gates pass, and each was re-run from a clean copy before this draft. The bundle ships the full certificate chains for

bundle file	sha256
MANIFEST.sha256 (root-of-trust)	dee28651ec9f5295d8c28d9d045e2d20 49753cc9f8e84ab0286dc6491fb8de91
THEOREM_MAP.md	824955ef552ca018f8b677fb16679b1d 271eb360be7f2caa0058d988f583f45b
THEOREM_MAP.json	46b411bd775e4c27016e59da76be1c0b 02e605650b1e452df93c065944e878ed
README.md	6a7c97566538afead4c785b4b862b9c2 8bc1d396514ea9d52ce879bb1bf8ef2d
check_bundle.sh	6abb8f2b67a825521f53953ec436e871 a85a5157c3ba0bdaf49eaf2987fd06e3

TABLE 4. Bundle front-page digests. Per-file digests for every certificate are in `MANIFEST.sha256`; the camera-ready PDF is deposited alongside the bundle and excluded from the bundle’s own manifest.

both bounds; the supplementary Euler-family saturation map, the negative/feasibility map (Sections 6, 8), and the GPU verification-economics study are summarized in the text but are not released as standalone certificate directories.

claim	bundle path
record manifest ($\Lambda \leq 0.197$): winding, stored-sums, sweep, tail, y -bridge legs + exact assembly	<code>record/record_package_197/</code>
record-binding theorem (two lines)	<code>record/binding/, record/binding_secondline/</code>
full stored-sums re-march	<code>record/fullmarch/</code>
winding second line (hash-chosen rectangles)	<code>record/winding_rects_secondline/</code>
sweep + tail second lines ($\times 3$)	<code>record/{sweep,tail,tail_row19}_secondline/</code>
bookkeeping lemmas (window gluing, $\times 2$ lines)	<code>record/nconst_glue/</code>
0.1965 chain (legs, assembly, sign-off)	<code>certified1965/</code>
binding-spine theorems: criterion (Thm 1.2), error budget (Thm 1.3), y -reduction	<code>record/criterion_theorem/, record/error_terms_audit/, certified1965/y_reduction/</code>
upstream artifact findings	<code>upstream/</code>
Lean implication skeleton, 0.197 chain (Sec. 7.5)	<code>lean/</code>

TABLE 5. Claim-to-artifact map for the frozen release bundle. The `record/`, `certified1965/`, and `upstream/` chains live under `bundle_21a/certificates/`; `lean/` is at the bundle root. Each listed verifier exits 0 (the three FLINT/Arb packages and the Lean build on their respective toolchains); per-file sha256 digests are in `MANIFEST.sha256` (Table 4).

Acknowledgements. This work runs entirely on open-source software, and the debt is real: the rigorous numerics rest on Arb/FLINT [Joh17], with independent lines in `mpmath` [J⁺23] and PARI/GP [The24]; and the entire enterprise builds on the Polymath15 project’s published methods, code, and committed artifacts [Pol19, Pol18] and on the Platt–Trudgian verification [PT21]. The prior-art computation of [[Dw20] deserves particular acknowledgement: it identified the exact barrier site and parameter class six years before this paper, in public, and asked for nothing in return.

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